Week 5 Online Class

Worksheet 3 (Q13 – 15)

Question 13. Prove that the sequence converges and find their limit.

$$(a_n) = \left(\sqrt{2}, \sqrt{2+\sqrt{2}}, \sqrt{2+\sqrt{2+\sqrt{2}}}, \ldots\right)$$

Solution. Note that $a_n > 0 \ \forall n \in \mathbb{N}$. Write

$$a_{n+1} = \sqrt{2+a_n} \ \forall n \in \mathbb{N} \ \text{ with } \ a_1 = \sqrt{2} \ \text{ and } \ a_2 = \sqrt{2+\sqrt{2}}\,.$$

(i) First, we prove that the sequence is bounded.

We prove that $a_n \geq \sqrt{2} \quad \forall n \in \mathbb{N}$.

It is clear that $a_1 = \sqrt{2}$.

Now assume that $a_k > \sqrt{2}$. Then

$$a_{k+1}^2 = 2 + a_k > 2 + \sqrt{2} > 2 \Rightarrow a_{k+1} > \sqrt{2}.$$

Hence, $a_n \geq \sqrt{2} \quad \forall n \in \mathbb{N}$.

Next, we prove that $a_n < 2 \quad \forall n \in \mathbb{N}$.

It is clear that $a_1 = \sqrt{2} < 2$.

Now assume that $a_k < 2$. Then

$$a_{k+1}^2 = 2 + a_k < 2 + 2 = 4 \Rightarrow a_{k+1} < 2.$$

Hence, $a_n < 2 \quad \forall n \in \mathbb{N}$.

Hence, $\sqrt{2} \leq a_n < 2 \quad \forall n \in \mathbb{N}$, the sequence is bounded.

(ii) Secondly, we show that the sequence is increasing.

Now

$$a_{n+1} - a_n = \sqrt{2 + a_n} - \sqrt{2 + a_{n-1}} = \frac{a_n - a_{n-1}}{\sqrt{2 + a_n} + \sqrt{2 + a_{n-1}}}.$$

By induction, if $a_n - a_{n-1} > 0$, then $a_{n+1} - a_n > 0$.

Now since $a_2 - a_1 > 0$.

So, we conclude that the sequence is increasing.

(iii) Apply MCT to conclude that the sequence converges.

Now, by MCT, the sequence converges because it is bounded and monotone.

(iv) Apply Algebraic Limit Theorem 6.1.

Let $\lim(a_n) = a \Rightarrow \lim(a_{n+1}) = a$.

Solving $a^2 = 2 + a$ we obtain $a = 2$ $(a > 0)$.

Question 14. Prove that the sequence $(a_n) = (x^n)$ converges if $-1 < x \leq 1$, where $\lim(a_n) = \begin{cases} 0, & |x| < 1 \\ 1, & x = 1 \end{cases}$; and the sequence diverges if otherwise.

Proof. Case 1. If $x = 1$ or $x = 0$,

then $(a_n) = (1)$ or $(a_n) = (0)$ are convergent

(constant) sequences which converge to 1 or 0.

Case 2. When $0 < x < 1$, then for all $n \in \mathbb{N}$,

$x^n - x^{n+1} = x^n\left(1 - x\right) > 0$ because $0 < x < 1$ and $x^n > 0$.

Now because $x > x^2 > x^3 > \cdots > x^n > \cdots > 0$,

the sequence $(a_n) = (x^n)$ is decreasing and bounded below.

Hence, by MCT, the sequence converges to $\lim(x^n) = a$.

To compute a, note that $(x^{2n}),(x^{2n+1})$ are subsequences

so by Theorem 8.3, $\lim(x^{2n}) = a = \lim(x^{2n+1})$.

Now, by Theorem 6.1(c),

$a = \lim(x^{2n}) = \lim(x^n) \times \lim(x^n) = a^2 \Rightarrow a = 0,1$; and

$a = \lim(x^{2n+1}) = \lim(x) \times \lim(x^n) \times \lim(x^n) = xa^2 \Rightarrow a(1 - xa) = 0$.

If $a = 1$, then $x = 1$, a contradiction. Hence, $a = 0$.

Altenative (Hardy). When $0 < x < 1$, write $x = \frac{1}{1+b}$, $b \in \mathbb{R}^+$.

Now $(1+b)^n \geq 1 + nb \ \forall n \in \mathbb{N}$ by the binomial theorem (this can be proved by induction). Since $nb \to \infty$ as $n \to \infty$, it follows that $(1+b)^n \to \infty$ and therefore, $x = \frac{1}{1+b} \to 0$.

Case 3. For $-1 < x < 0$, we claim that $\lim(x^n) = 0$.

We want to show this (by using the definition):

$\forall \varepsilon > 0, \exists N \in \mathbb{N} \ni \left|x^n - 0\right| < \varepsilon$ for all $n > N$,

where $-1 < x < 0$).

But this is equivalent to $\left|x^n - 0\right| < \varepsilon$ for $0 < x < 1$.

Hence, by Case 2, $\lim(x^n) = 0$.

Case 4. For $x > 1$ or $x < -1$, $(a_n) = (x^n)$ is unbounded and therefore divergent.

Case 5. For $x = -1$, $(a_n) = ((-1)^n)$ is divergent.

Question 15. Let $\left(a_n\right)$ and $\left(b_n\right)$ be convergent sequences and sequence $\left(c_n\right)$ be defined as $c_n = \left|a_n - b_n\right| \forall n \in \mathbb{N}$. Prove that $\left(c_n\right)$ is convergent.

Proof. Since $\left(a_n\right)$ and $\left(b_n\right)$ are convergent, it follows that they are Cauchy, i.e.,

$$\begin{cases} \forall \varepsilon > 0, \exists M \in \mathbb{N} \ni \left|a_n - a_m\right| < \frac{\varepsilon}{2} \forall m, n > M \\ \forall \varepsilon > 0, \exists N \in \mathbb{N} \ni \left|b_n - b_m\right| < \frac{\varepsilon}{2} \forall m, n > N \end{cases}.$$

Therefore, for $m, n > \max\{N, M\}$, it implies that

$$\left|c_n - c_m\right| = \left|\left|a_n - b_n\right| - \left|a_m - b_m\right|\right| \leq \left|\left(a_n - b_n\right) - \left(a_m - b_m\right)\right| \leq \left|a_n - a_m\right| + \left|b_n - b_m\right| < \varepsilon,$$

which mean $\left(c_n\right)$ is Cauchy and hence it is convergent.